A Tactical Strategy using ETFs: Harvesting Volatility Risk Premia & Crisis Alpha

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1 Introduction

I discuss a systematic approach to investing in volatility risk premia through ETFs. The reason behind ETFs is mostly due to my personal interest in volatility investing and being able to manage the risk of the strategy on my own without any capital constraint. However, I will be using futures contracts mostly to backfill the data for my backtest but the main idea is focused around futures contracts whose exposure can be obtained through ETFs.

I first discuss the main methodology. My primary model is based on a short volatility product. Several of the short volatility ETFs have suffered terrible losses due to a number of market turbulence. These periods of turbulence can be classified as bear markets with higher volatility of volatility and explosion of negative convexity (COVID 19 for example), or events related to liquidity concerns, like periods when there are significant negative gamma positions (buy high/ sell low) in market, caused primarily due to dealer hedging pressures (Volmageddon for example) and rebalancing triggers for vol targeted/passive funds' deleveraging purposes. Thus, the key for my research was to investigate whether ex-ante there are market fundamentals which can potentially inform me of the direction of volatility over the short term. The signal I propose is the average of the slopes calculated from Cash VIX term structure and VIX Futures contracts. The expectation of the systematic process is to stay in short volatility position in most of the periods while tactically cutting risk or moving to long volatility asset in periods of turbulence in order to avoid potential abrupt losses. During normal periods, the term structure of expected volatility tends to be upward sloping, or stays in contango. This is due to the expectation of higher volatility in the future which increases the uncertainty around hedging costs (lower expected spread between future realized volatility and implied volatility at trade initiation). This presents an evidence of positive volatility risk premium. Prior to heightened market volatility, which is caused mostly due to the above mentioned reasons, the term structures of implied volatility and its expectation show signs of flattening relative to normal periods. Thus ex-post we move into a regime where the short term expected volatility is higher than long term expected volatility, or in other words backwardation. The corresponding sell-offs cause the volatility insurance claim to be realised by long volatility investors.

My primary model is based on the slope measure defined above. Conditional on the primary model signal, which takes advantage of the mean reversion nature of volatility, I tactically rotate (rebalanced end of each day) between a short volatility ETF or a long volatility ETF, each scaled to 20% volatility. However, the losses from short/long volatility exposure during a false positive scenario can deteriorate the risk adjusted return of the strategy. Thus, instead of keeping my unallocated capital in cash, I allocate them to a 15% volatility trend following strategy (internal program rebalanced daily) based on selected commodity and bond ETFs. The reason behind selecting only commodities and bonds as asset classes is to minimize the correlation against my volatility strategy, which has an equity bias as well. The distribution of volatility carry strategies are of convergent nature, also known as negative skew. Trend following strategies on the other hand, essentially bear a long gamma profile with divergent distribution, thus are expected to provide cushion during periods of false positives generated by the tactical volatility signal.

Furthermore, in order to augment my primary model and improve its overall hit rate, I also propose a meta model for the primary volatility strategy. I estimate a regression model to quantify the probability of my strategy failure or in other words the probability of a successful bet. The conditional probabilities are based on variables related to the term structure of both cash VIX and futures, i.e. essentially several individual transformations derived from the term structures which have ex-ante information on the direction of the slope, other than the main measure described above. The out of sample conditional probabilities are then used to position size my volatility strategy, i.e. when the conditional probability is high, there is a higher chance of a correct bet. I find that the meta strategy, alongside the trend following buffer provided superior risk adjusted returns relative to the individual components during the sample period in question.

2 Data

Table 1 lists the assets I used for the backtest. VRP stands for Volatility Risk Premia. For each strategy I list the ETFs used and its underlying leverage. I use a mix of levered and unlevered ETFs for capital efficiency purposes. I also list the futures contract for each asset which is primarily used for backfilling the data as far as possible. The assets for the CTA starts from 1990 onwards and the VRP assets start from 2006 onwards.

Table 1: List of Assets

VRP			**CTA**		
ETF Leverage	**ETF**	**Futures**	**ETF Leverage**	**ETF**	**Futures**
-1x	SVIX/SVOL	UX1	2x	UGL	GC1
1x	VIXY	UX1	1x	USO	CL1
			1x	USL	CL12
			1x	UNG	NG1
			1x	SOYB	S 1
			1x	JJC	HG1
			3x	TMF	US1
			3x	TYD	TY1

The table below outlines the list of assets used for constructing signals and variables for the main strategy and its meta model. For the VRP strategy, I use three sets of datatypes, cash VIX, futures and open interest information for VIX options. The signals for CTA are all price based measures, and thus the assets themselves are sufficient. All my returns are excess of cash rate (based on 1M Libor).

Table 2: Data Used for VRP Signals

Cash	**Futures**	**Open Interest**
VIX Index	UX1	VIX Put
VIX3M Index	UX2	VIX Call
VIX6M Index	UX3	
VIX1Y Index	UX4	
	UX5	
	UX6	
	UX7	
	UX7	
	UX8	
	UX9	

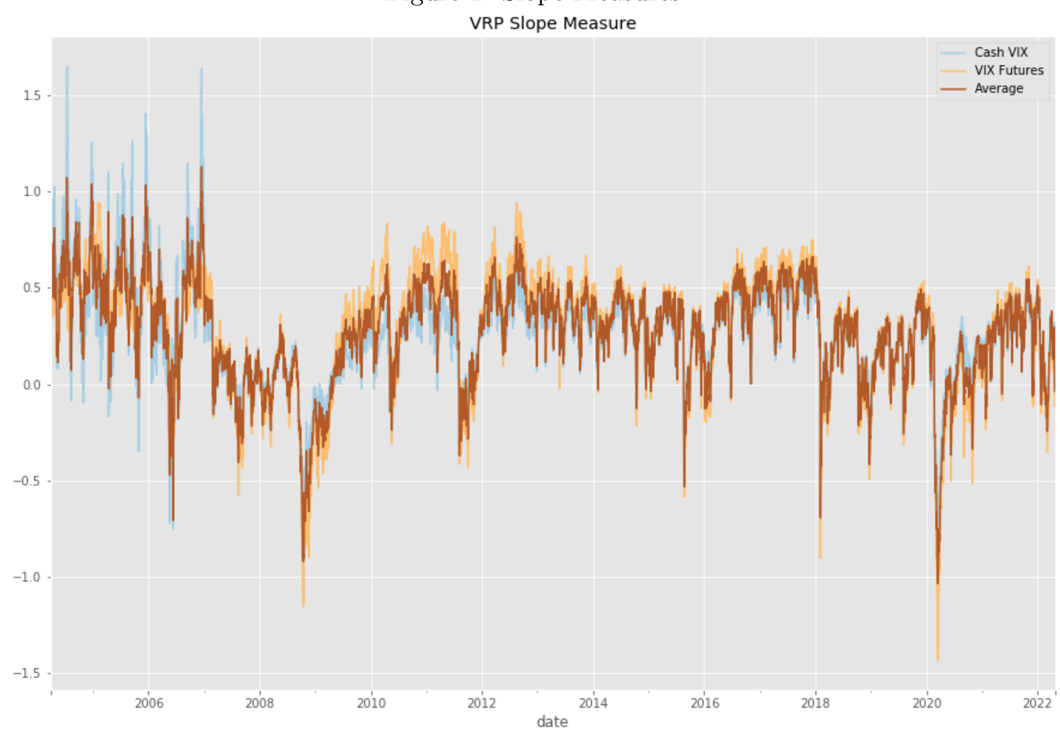
3 Primary Model - Volatility Risk Premium

My primary signal, denoted as $\hat{\lambda}_t$ is computed as the average slope between cash VIX (VIX) and futures ($VIXF$). The slope is estimated for each available maturity at each point in time through a simple cross sectional regression.

$$\hat{\lambda}_t = 0.5 * (\lambda_{VIX,t} + \lambda_{VIXF,t}) \tag{1}$$

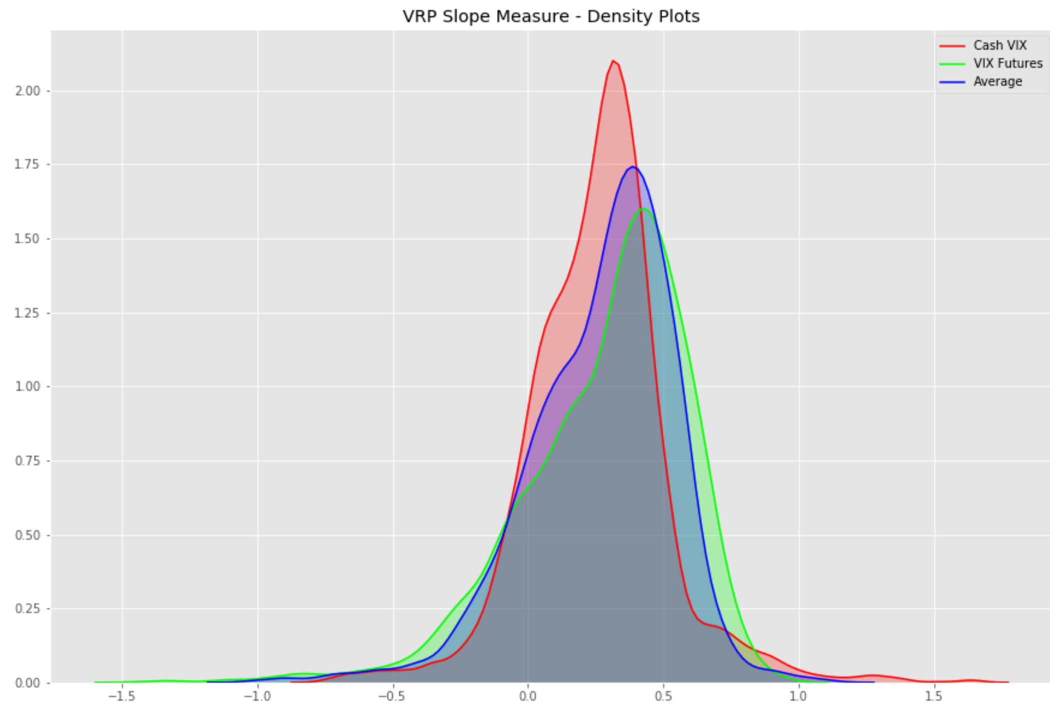
$$V^{\tau}_{X,t} = \alpha + \lambda_{X,t}\tau + \epsilon^{\tau}_t \ ; X \in (VIX, VIXF) \tag{2}$$

Figure 1: Slope Measures



The full sample rank correlation between the two measures is 0.87, thus suggesting positive and higher degree of co-movement. The unconditional average slopes for VIX and VIXF are 0.26 and 0.29 respectively. The figure above plots the time series of the two measures plus their average, which is the final indicator. One observation clearly stands out that the slopes turns negative during periods of negative market shocks, with all three measures dipping below -0.50 during GFC, Volamageddon and COVID meltdown. Analysing the density plots, the VIX slope is slightly positively skewed with much lesser fat tails, where as the VIXF slope distribution shows relatively more dispersion with more negative skew. Averaging the two measure yields a much less variant and less skewed distribution.

Figure 2: Slope Measures



Next, I also compute a z-score for my slope measure, $Z_{\hat{\lambda},t}$.

$$Z_{\hat{\lambda},t} = \frac{\hat{\lambda}_t - EWMA_{t-1}\left[\hat{\lambda}; span = 63\right]}{EWSD_{t-1}\left[\hat{\lambda}; span = 63\right]} \tag{3}$$

My long-short trading signal (here long implies going long on the SVOL, where the underlying position is short volatility) is a combination of the sign of the slope and the deviation of the z-score based on a threshold (which I set to 1 standard deviation).

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$$X_{t+1}^{VRP} = \begin{cases} 1; & if\ (\hat{\lambda}_t\ > 0\ \&\ Z_{\hat{\lambda},t}\ >\ 0) \\ -1; & if\ (\hat{\lambda}_t\ > 0\ \&\ Z_{\hat{\lambda},t}\ <\ 0) \\ -1; & if\ (\hat{\lambda_{t-1}}\ > 0\ \&\ Z_{\hat{\lambda},t-1}\ <\ 0)\ \&\ (\hat{\lambda}_t\ < 0\ \&\ Z_{\hat{\lambda},t}\ <\ -1) \\ 1; & if\ (\hat{\lambda_{t-1}}\ < 0\ \&\ Z_{\hat{\lambda},t-1}\ <\ -1)\ \&\ (\hat{\lambda}_t\ < 0\ \&\ Z_{\hat{\lambda},t}\ >\ -1) \end{cases} \tag{4}$$

Let's go over the four cases of the strategy. The first case implies a positive slope accompanied by positive z-score, which denotes an upward trend in the measure. This is the ideal scenario to stay long SVOL. The second case implies a positive slope with a negative breakout; suggesting gradual flattening of the term structure, this signals closing out position in SVOL and moving to VIXY, which is the long volatility ETF. Next, conditional on the second case at $t-1$, if the current slope is negative with a downward trend breakout greater than 1 standard deviation, this will suggest holding VIXY, since the persistence in volatility is high and can take longer than the threshold to normalize. Finally, conditional on case 4, if there is an upward breakout trend in the negative slope of greater than -1 standard deviation, this implies gradual increase in slope or gradual reversion to low volatility regime and thus moving the position back to SVOL.

The daily return of the strategy can be written as follows,

$$rx_{VRP,t+1} = \left[\frac{X_{t+1}^{VRP}+1}{2}\right]\frac{20\%}{\sigma_{SVOL,t}}rx_{SVOL,t+1} + \left(1-\left[\frac{X_{t+1}^{VRP}+1}{2}\right]\right)\frac{20\%}{\sigma_{VIXY,t}}rx_{VIXY,t+1} \tag{5}$$

The ex-ante volatility is estimated using a exponentially weighted process with a center of mass of 63 days.

4 Primary Model Buffer - Trend Following System

Now, I outline the methodology behind the Trend Following system. The primary intention of this program is to allocate the excess capital as an augmentation to the tactical strategy and boost long gamma exposure (or long straddle) while providing buffer against possible return decay due to mistimed volatility exposure. For each CTA asset i, the signal strength is calculated based on the median of three separate models.

$$X_{i,t}^{CTA} = med\left(S_{i,t}^{k}\right)\ ;\ X_{i,t}^{CTA} \in [-1,1] \tag{6}$$

Each model k is processed from 3 different trend estimators, denoted as S^k,

1. Risk Adjusted Momentum:

For each lookback $n \in [21, 63, 252]$, the risk adjusted momentum signal is calculated as,

$$x_n = \frac{\Pi_t^n(1+rx_t)}{\sigma_n\sqrt{n}} \tag{7}$$

The numerator of the formula is simply the compound returns over the period. $\sigma_n\sqrt{n}$ is the daily volatility scaled over the lookback period.

2. EMA Crossover:

For each lookback set, $s \in [5, 10, 20]$ & $n \in [20, 40, 80]$, the EMA Crossover signal is calculated as,

$$x_n = \frac{EWMA\left[P; hl[s]\right] - EWMA\left[P; hl[p]\right]}{EWSD\left[P; hl[p]\right]} \tag{8}$$

$$hl[s] = \log 0.5/\log\left(1-1/s\right) \tag{9}$$

3. EMA Breakout:

For each lookback $n \in [21, 63, 252]$, the EMA Breakout signal is calculated as,

$$x_n = \frac{P - EWMA\left[P; span=n\right]}{EWSD\left[P;; span=n\right]}. \tag{10}$$

Next, each x_n is processed into a signal strength as follows,

- For each model and each lookback n, I normalize each x_n by their exponential weighted volatility.

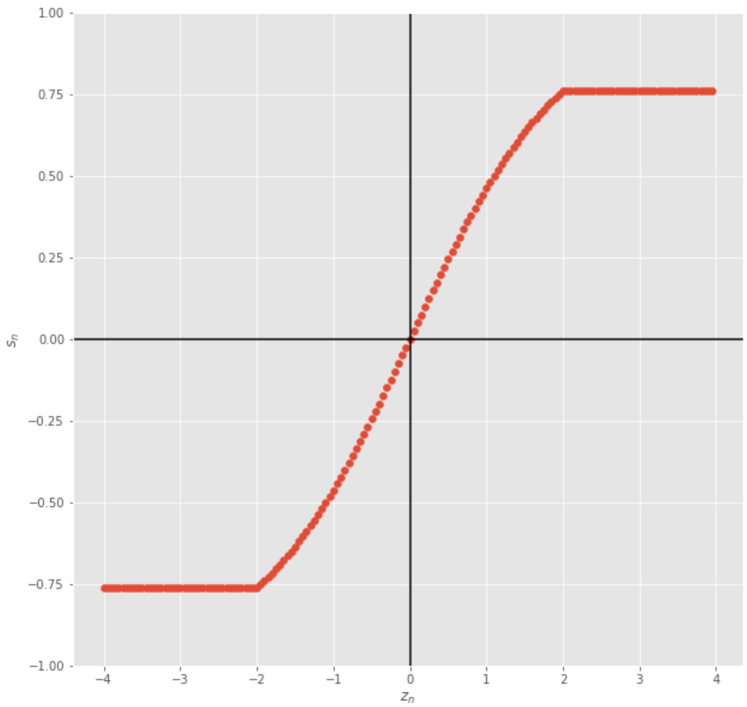
$$z_n = \frac{x_n}{EWSD[x_n, span = 252]} \tag{11}$$

- Next, each z_n is transformed into a position size based on a sigmoid function capped at 2 sigma, denoted as s_n,

$$s_n = 2 * \left[\min\left(\max\left(\frac{1}{1+e^{-z_n}}, 1 - \frac{1}{1+e^{-2}}\right), \frac{1}{1+e^{-2}}\right)\right] - 1 \tag{12}$$

Figure 3 below shows an illustration of the formula above,

Figure 3: Long Short Position Sizing with 2 Sigma Cap



- The final signal strength from each model k is computed as,

$$S^k = \sum_n w_n s_n \tag{13}$$

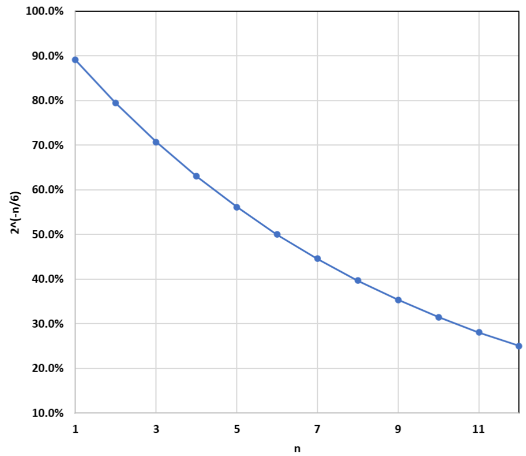
$$w_n = \frac{2^{-n/6}}{\sum_n 2^{-n/6}} \tag{14}$$

Here, the raw weights, denoted as $2^{-n/6}$, is chosen based on a halflife of 6 months. Figure 4 demonstrates an illustration of the weight decay under various lookbacks,

The trend following strategy return is computed as follows,

$$rx_{t+1}^{CTA} = \frac{15\%}{\bar{\sigma}_t} \sum_{i=1}^{N} w_{i,t}^{gross} * X_{i,t}^{CTA} * \left(\frac{15\%}{\sigma_{i,t}}\right)^2 rx_{i,t+1} \tag{15}$$

Figure 4: Raw Weights as a Function of Lookback - 6 Month Halflife



$$\bar{\sigma_t} = \sqrt{\sum_{q=t_0}^{t} \frac{\omega_q^{net\prime} \Sigma_q \omega_q^{net}}{t - t_0}} \tag{16}$$

Here, ω_t^{net} is a vector of net weights where each row is equal to $w_{i,t}^{gross} * X_{i,t}^{CTA} * \left(\frac{15\%}{\sigma_{i,t}}\right)^2$. Using the variance ratio $\left(\frac{15\%}{\sigma_{i,t}}\right)^2$ instead of volatility, is equivalent to an inverse variance portfolio within each cluster. Here, $\sigma_{i,t}$ is the annualized volatilty from the covariance matrix Σ_t. $\bar{\sigma_t}$ is the long term average of the instantaneous portfolio volatility computed over an expanding window. The covariance matrix is estimated from volatilities with 60 day halflife and a correlation matrix based on 252 day halflife. The gross weights for each commodity is 10% and 20% for the fixed income assets. This ensures a notional allocation of 60% in commodities and 40% in fixed income. The underweight in fixed income is primarily due to the lack of breadth, since I only have two assets in this cluster.

Now, allocation between the CTA and VRP is quite simple. For example, if 40% notional weight is allocated to SVOL, under that case I will allocate 60% of my remaining capital to the CTA Program. One thing to note that within the CTA program, certain assets may need to be shorted based on their signal strength, in such cases a similar exposure could have also been obtained through an inverse ETF (like the VIX), thus avoiding the hurdles caused due to borrowing cost and margin requirement.

Table 3 shows the Full Sample Performance of the 15% Volatility CTA,

Table 3: CTA Full Sample Performance - 1990-Present

CTA @ 15% Vol	
Arith. Mean	11.6%
Compound Mean	10.6%
Volatility	16.3%
SR	0.71
t-Stat	4.07
Geo SR	0.65
Max DD	25.2%
Ulcer	1.17
Calmar	0.46
Skewness	0.07
Kurtosis	9.02

5 Meta Model - Volatility Risk Premium

Next, I describe the meta model for VRP strategy. The goal of the model is to act as a risk management overlay to predict unfavorable regimes for the strategy. The objective is to estimate the probability of a correct bet for each trade and then based on a given threshold of probability, either go on with the trade as prescribed by the primary model or vice versa. The meta model is designed to reduce model uncertainty and the false positive generated from the primary strategy. The conditional probability is estimated based on a linear ridge regression. Here I describe the conditioning variables briefly,

- Roll Yield:

The VIX roll yield (RY_t) is computed as,

$$RY_t = \frac{UX1_t}{VIX_t} \tag{17}$$

- VIXF slopes at different points of the curve:

These variables are simply different combination of ratios between the different points of VIXF term structure, to capture all the potential variations not captured by the primary slope measure. The several combinations are as follows,

$$\lambda_{VIXF1:VIXF2,t} = \frac{UX2_t}{UX1_t} \tag{18}$$

$$\lambda_{VIXF1:VIXF3,t} = \frac{UX3_t}{UX1_t} \tag{19}$$

$$\lambda_{VIXF4:VIXF7,t} = \frac{UX7_t}{UX4_t} \tag{20}$$

$$\lambda_{VIXF1:VIXF9,t} = \frac{UX9_t}{UX1_t} \tag{21}$$

- VIX slopes at different points of the curve:

Same as the futures, I compute several slope measures, this time using the Cash VIX term structure. The several combinations are as follows,

$$\lambda_{VIX1:VIX3,t} = \frac{VIX3M_t}{VIX_t} \tag{22}$$

$$\lambda_{VIX1:VIX6,t} = \frac{VIX6M_t}{VIX_t} \tag{23}$$

$$\lambda_{VIX1:VIX12,t} = \frac{VIX1Y_t}{VIX_t} \tag{24}$$

$$\lambda_{VIX3:VIX6,t} = \frac{VIX6M_t}{VIX3M_t} \tag{25}$$

- Put/Call:

I also incorporate one volume based information, the put-call ratio of VIX options. Unlike the traditional put-call ratios, which incorporate volume based information. I use open interest to compute the ratio. Using open interest instead of volume puts the focus only on active contracts.

I transform each of the variables into a breakout measure, similar to the CTA breakout, using a weighted lookbacks of 63 and 190 days. This gives a total of 9 initial set of variables for the meta model. Next, for each raw variable related to slope (Put/Call ratio being the exception), I compute a dummy variable which combines the information from the raw slope variables and the breakout measures computed earlier, essentially an indicator for a positive slope regime ex-ante. For each slope variable, I compute D_t as follows,

$$D_t = \begin{cases} 1; & if\ (\lambda_{\tau_1:\tau_2,t}\ > 0)\ \&\ (EMABreakout_{\lambda_{\tau_1:\tau_2},t})\ >\ 0 \\ 0; & Otherwise \end{cases} \tag{26}$$

This dummy informs about the current state of the particular slope of the term structure. The movement of the different points of the term structure are not strictly upward or downward sloping and so in certain cases there maybe potential changes in local slopes which can be informative ex-ante to explain the unexplained variation from the primary model. Combining the 9 breakout measures with the 8 local slope regime indicators, we come to the final set of variables for our meta model.

I define the meta model as follows,

$$\Phi(y_{t+1})^{-1} = \Gamma' X_t + \epsilon_t \tag{27}$$

Here, $\Phi(.)^{-1}$ is the standard normal inverse function; $y_{t+1} = 1$, if $X^{VRP}_{t+1} = sign(rx_{SVOL,t+1})$, and 0 otherwise; Γ is a vector of regression coefficients.

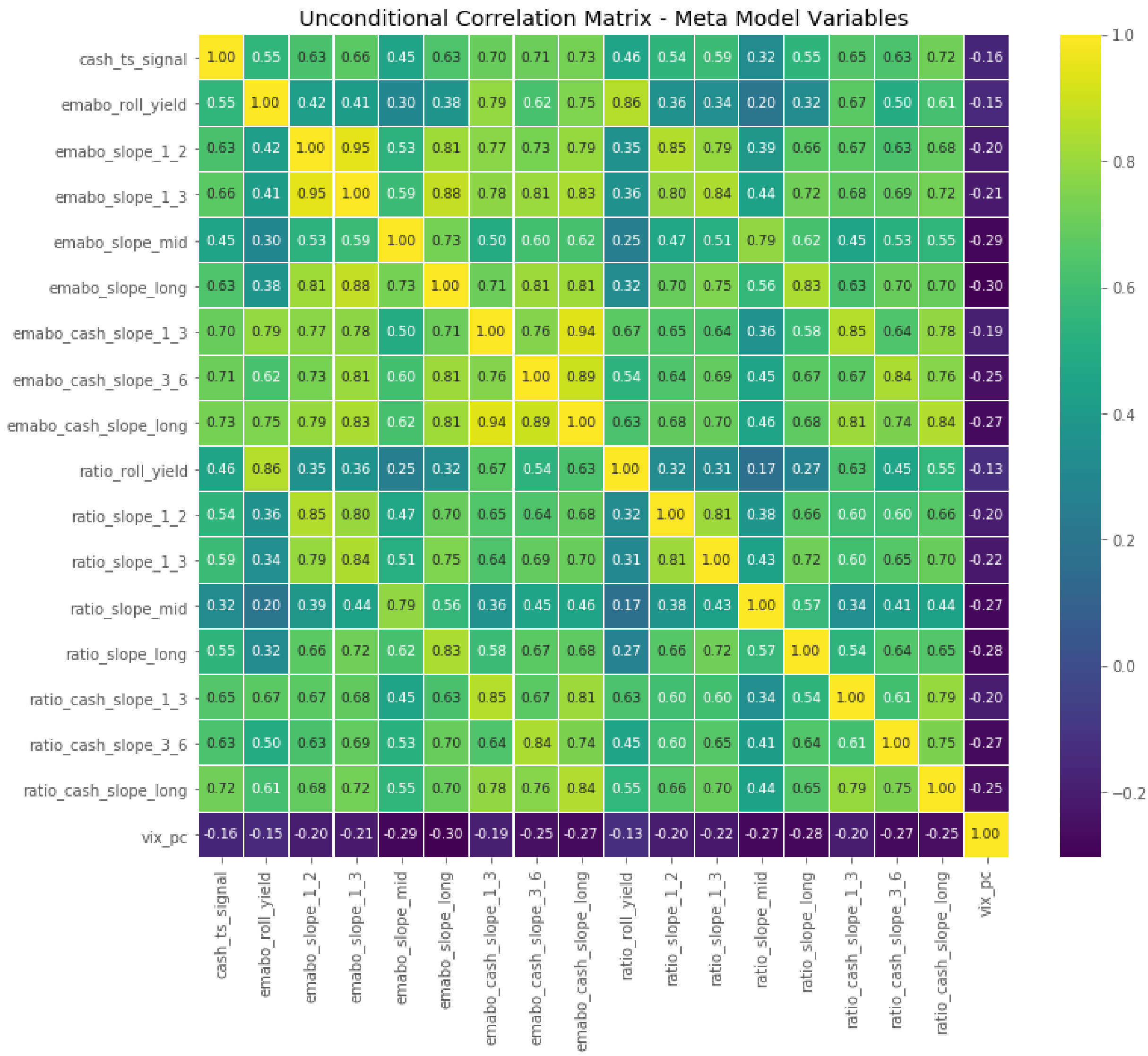
The ridge regression estimator can be written as follows,

$$\Gamma = (\mathbf{X}'\mathbf{X} + \alpha\mathbf{I})^{-1}\mathbf{X}'\mathbf{z} \tag{28}$$

$\mathbf{X}$ is a matrix of conditioning variables, $\mathbf{I}$ is an p dimensional identity matrix with p being the number of variables, $\mathbf{z}$ is a vector of the dependent variable within the in-sample period and α is the ridge parameter to reduce out of sample overfitting with $\alpha > 0$. I estimate the model over an expanding window, starting with 2 years of data from 2007/01. At a given point in time the estimated regression coefficients are used to estimate the probabilities for the next 21 days. After 21 days, I add those data points and re-calibrate my model. To calibrate α within the in-sample expanding window, I apply a time series cross validation with a grid search procedure to find the optimal α at each calibration period. The out-of-sample predicted probabilities $\Phi(y_{t+1})$ are computed by applying the cumulative distribution function on the regression prediction.

Next, I pick 40% as a threshold for the predicted probabilities, over which I have confidence on the primary

Figure 5: Meta Model Variables - Rank Correlation Matrix



model signal. The probability adjusted signal is simply computed as,

$$X_{t+1}^{VRP\ Meta} = 1_{\Phi(y_{t+1})>0.40} * X_{t+1}^{VRP} \tag{29}$$

As per equation 5, in the case when $X_{t+1}^{VRP\ Meta} = 0$, the model will allocate 50% to both SVOL and VIXY, simply leaving to random chance.

6 Performance Analysis

The backtest starts from 2008/12 till present. Each strategy is daily rebalanced, with results computed absent of transaction cost. Figure 6 shows the table for the performance statistics. VRP, CTA and VRP Meta are the individual volatility, trend following and volatility meta strategy respectively. VRP + CTA is the combo strategy where the unused capital is allocated to CTA. VRP + CTA Meta is essentially the same as VRP + CTA, except the VRP signal is adjusted for the meta model. The standalone VRP strategy performed with a Geometric Sharpe of 0.71. Augmenting VRP with the meta model yields an improved performance, with Geometric Sharpe of 1.01. The simple VRP + CTA strat outperformed the standalone VRP and CTA, generating relatively higher compound return throughout the period. However, the VRP + CTA meta strategy stands out to be best performer, with a much higher geometric sharpe of 1.19. Furthermore, comparing their performance against traditional assets, ES1, TY1 and SVOL, we see all these strategies have outperformed on a risk adjusted basis during this period. The VRP signal has much lower maximum drawdown and positive skew compared to the standalone SVOL strategy.

Figure 6: Performance Statistics: 2008/12-Present

	VRP	CTA	VRP Meta	VRP+CTA	VRP+CTA Meta	ES1	TY1	SVOL
Arith. Mean	15.8%	10.0%	15.9%	22.2%	23.4%	14.1%	5.9%	48.5%
Compound Mean	14.6%	9.2%	15.8%	21.4%	23.8%	13.4%	3.7%	28.3%
Volatility	20.6%	15.7%	15.7%	24.1%	20.0%	17.8%	21.0%	67.0%
SR	0.77	0.64	1.01	0.92	1.17	0.79	0.28	0.72
t-Stat	2.85	2.38	3.77	3.43	4.34	2.96	1.04	2.70
Geo SR	**0.71**	**0.59**	**1.01**	**0.89**	**1.19**	**0.75**	**0.18**	**0.42**
Max DD	28.3%	23.3%	26.3%	37.7%	31.1%	34.2%	42.3%	92.3%
Ulcer	1.53	1.17	1.90	1.80	2.40	2.56	0.39	1.12
Calmar	0.56	0.43	0.61	0.59	0.75	0.41	0.14	0.53
Skewness	0.87	0.75	0.49	1.02	0.95	-0.33	0.26	-1.54
Kurtosis	8.56	18.51	17.08	11.10	20.07	11.07	11.86	8.07

Figure 7: Geometric Sharpe Decay by Signal Lag

Signal Lag	VRP	VRP Meta	VRP+CTA	VRP+CTA Meta	ES1	TY1	SVOL
1 Day	0.71	1.01	0.89	1.19			
3 Day	0.36	0.69	0.54	0.86			
5 Day	0.48	0.77	0.68	0.94	**0.75**	**0.17**	**0.42**
10 Day	0.43	0.44	0.56	0.60			
21 Day	-0.12	0.29	0.12	0.52			

Next, I do a robustness check to assess the speed of the signal and performance decay. Figure 7 shows the Geometric Sharpe ratios for each strategy with four different signal lags, 1, 3, 5, 10, 21 days. I also show the Geo SRs for the traditional assets for comparison. One key takeaway is that the VRP + CTA Meta strategy holds decent outperformance on a risk adjusted basis, even upto 5 day lags. The primary VRP strategy performance decays faster than its Meta counterpart with higher lags, with a SR of -0.12 at 21 day lag. Figure 8 and 9 shows the performance curves in log space, one for the different strategy combos, and the other one with the VRP + CTA Meta strategy against the traditional assets; all scaled to unconditional 10% volatility.

Figure 8: Performance Curve

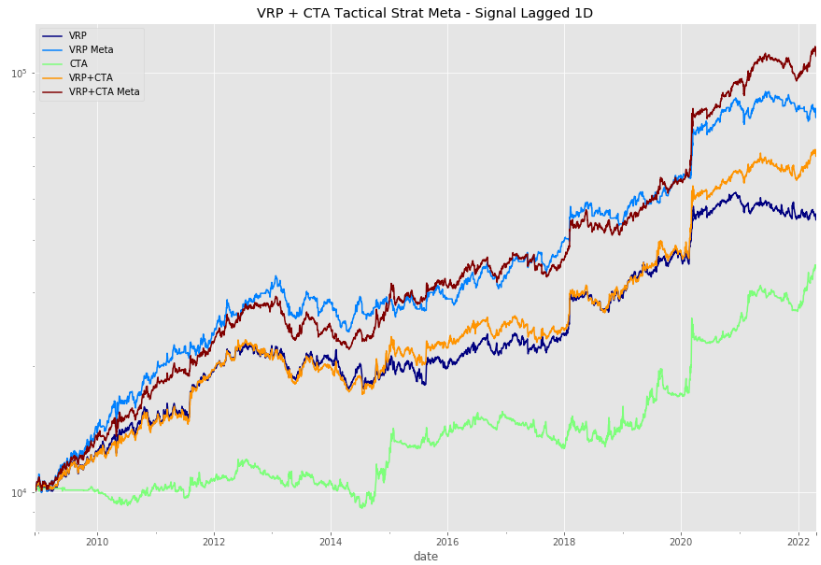


Figure 9: Performance Curve against Traditional Assets

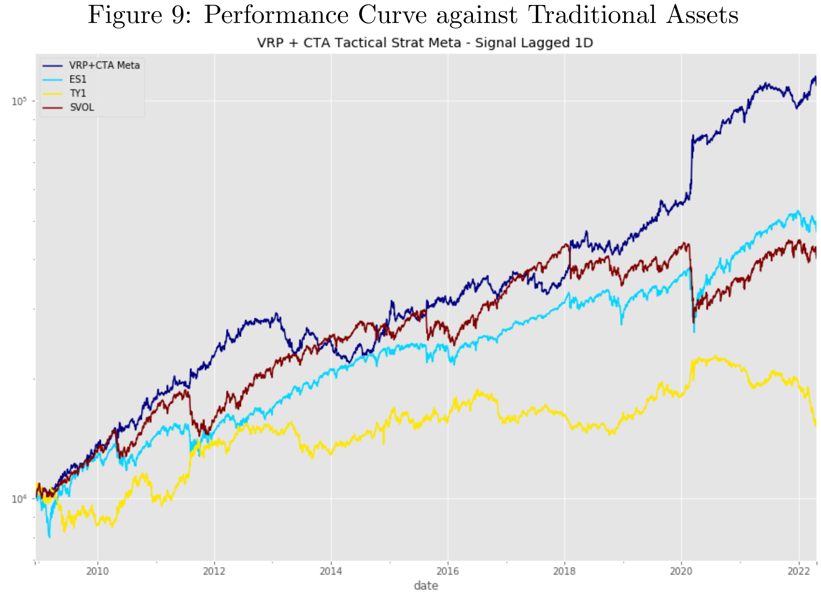


Figure 10: Strategies & Assets - Rank Correlation Matrix (2008/12-Present)

Next, in Figure 10, I show the rank correlation matrix for the different strategies and assets. The CTA shows negative correlation to ES1 and SVOL, thus acting as a good diversifier. One interesting takeaway is the correlation of the VRP + CTA Meta strategy against the traditional assets. The strategy has close to zero correlation to TY1, with strong positive rank correlation with CTA, ES1 and SVOL. However, the relationship is strongly non-linear as the linear correlation measure is much lower (0.07 against ES1 and 0.27 against SVOL). This is also verified by the regression plot in Figure 11. In Figure 12, I also plot the exponentially weighted conditional pairwise correlations for VRP + CTA Meta against the traditional assets. There is significant time variation in dependence against the traditional assets, with strong positive correlation with bonds during crisis periods and negative correlation against SVOL and ES1.

Figure 11: Regression Plot

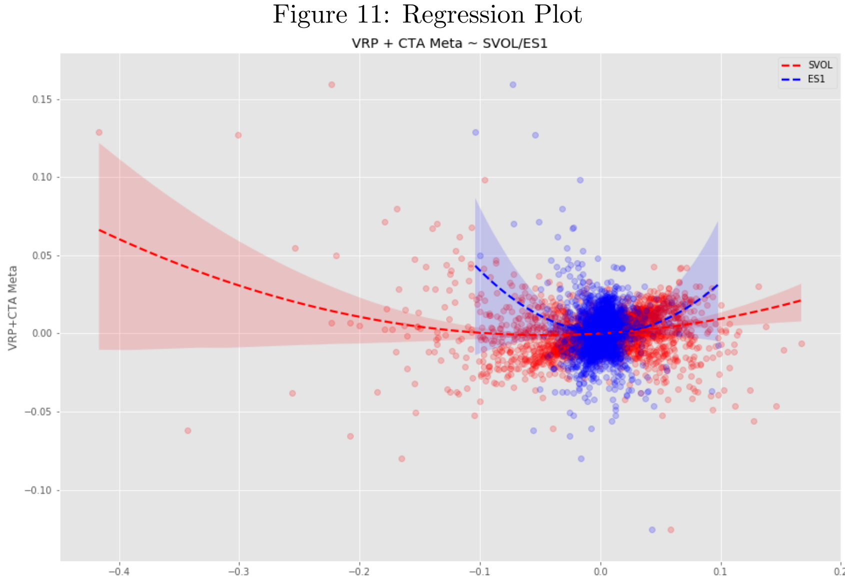
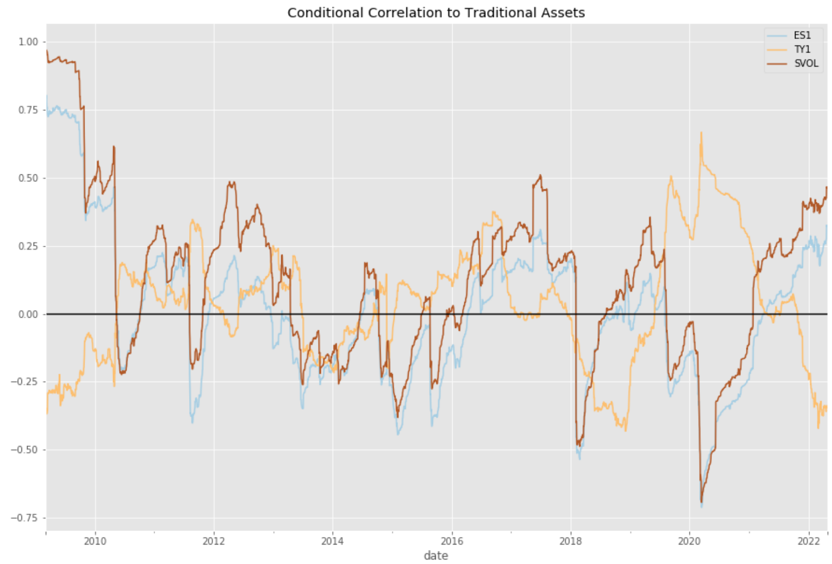


Figure 12: Conditional Correlation: VRP + CTA Meta against ES1 and TY1



Next, I conduct spanning tests by simply running linear regression on each strategy against the benchmarks, in this case the traditional assets and CTA. The conditional variation for the standalone VRP model is quite low, relative to the VRP+CTA Meta Strategy. All models have statistically significant loading on CTA and SVOL. Also, all four combos demonstrated positive and statistically significant alpha throughout the period. The VRP + CTA Meta strategy has insignificant conditional loading to bonds and equities, with strong t-stats of 44.5 and 10.5 respectively on CTA and SVOL. This suggest that unlike the primary VRP strategy, The VRP + CTA Meta is able to preserve the characteristics of a short volatility strategy, while at the same time providing crisis alpha generated through CTA and long volatility positions.

Figure 13: Spanning Test - OLS Results (2008/12-Present)

	CTA	SVOL	TY1	ES1	Alpha Ann. (scaled to 10% Volatility)	Rsquared
VRP	**0.12**	**-0.08**	0.00	0.08	**8.51%**	*5.6%*
VRP Meta	**0.04**	**0.11**	0.00	-0.04	**8.12%**	*6.0%*
VRP+CTA	**0.76**	**-0.07**	0.01	0.05	**8.52%**	*30.4%*
VRP+CTA Meta	**0.78**	**0.11**	0.00	-0.06	**8.23%**	*37.6%*

Bolded coefficients |t-Stats| > 2.5

The next few plots show the model weights for the CTA and the overall VRP + CTA Meta Strategy.

Figure 14: Commodity CTA Weights

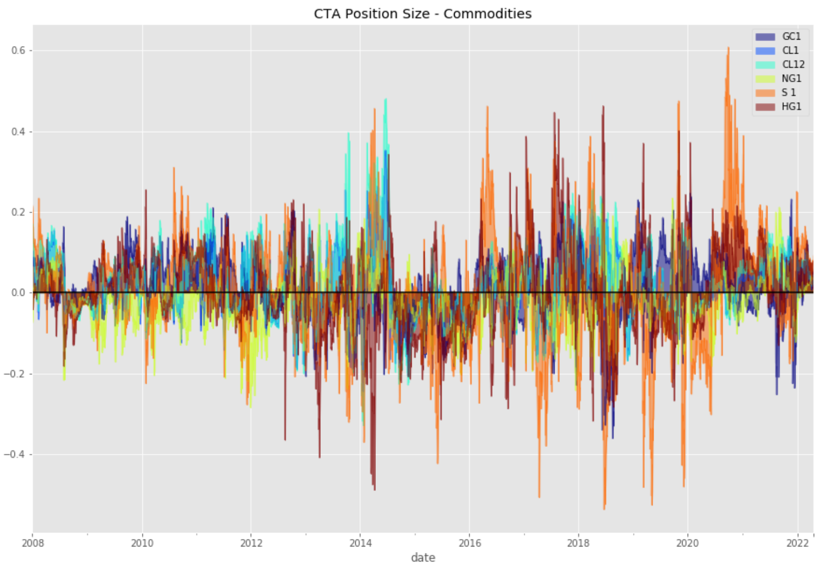


Figure 15: Fixed Income CTA Weights

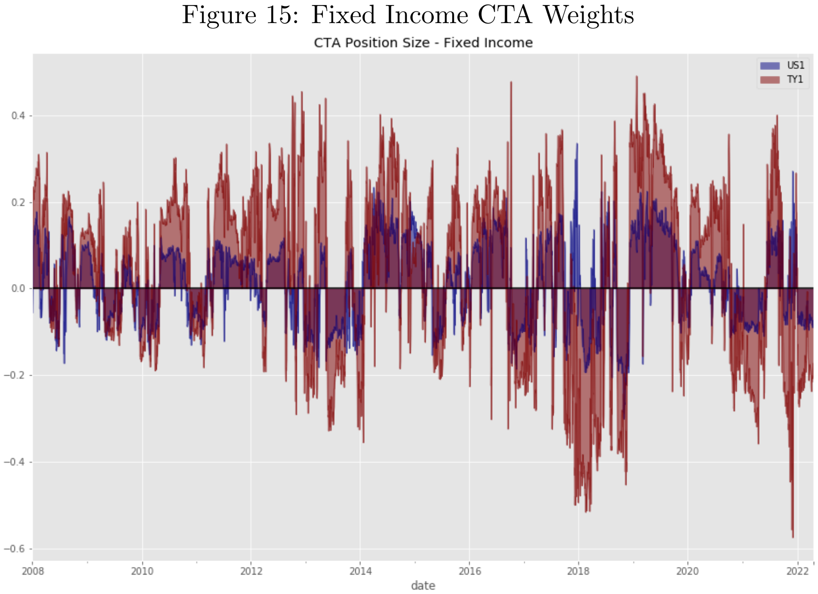


Figure 16: Meta Strategy Allocation